

Public datasets

RRRM-2 / OSD-771
age × arm × flight
kidney RNA-seq

OSD-513
kidney flight
RNA-seq

OSD-253
strain context
RNA-seq

OSD-462 / RR-10
multi-omic anchor
RNA + protein + phospho

**RNA state
recurrence**

Cross-cohort RNA contrast vectors
matrix-high / DCT-low remodeling direction recurs
RRRM-2 ISS-T ↔ OSD-513 (co-
sine 0.64) ↔ OSD-462 (cosine 0.87)

**OSD-462
multi-omic
anchor**

RNA recurrence
gate **PASS**

Protein abundance
concordance **NULL**

Phospho activity
NCC/SPAK regulatory
sites **suppressed**

Network translation
(LIONESS/node2vec)
NULL

**Regulator
activity**

KSEA kinase activity
WNK & SPAK/OSR1 inferred
activity down (positive control)

TF / pathway activity
decoupler PROGENy
+ DoRothEA
candidate regulator nomination

**Phenotype
anchor**

Phenotype anchoring — animal-matched within RR-10
DCT transport RNA program vs measured
NCC/SPAK regulatory phosphorylation
group-level concordant; animal-
level link suggestive (underpowered)

Evidence ladder RNA state → phospho activity (measured) →
DCT morphology (Siew 2024, literature) → physiology (future work)